

EcoPassEU: A Privacy-Aware Digital Product Passport DApp for Sustainable Consumer Goods in Europe

Fuyu Cao¹, Luxiao Cao¹, Detai Dong¹, Akshansh Rajora¹, and Tianwei Yu¹

University of Bristol, Bristol, UK

{1d25515, ne25392, mz25325, tk25240, np25024}@bristol.ac.uk

1 Introduction

Every year, billions of products enter the European market with sustainability labels, yet consumers have little reliable way of checking whether claims such as “low carbon” or “ethically sourced” are accurate. A study commissioned by the European Commission found that more than half of the green environmental declarations lack verifiable evidence to support, eroding public confidence in corporate environmental commitments [18].

To address this credibility gap, the Ecodesign for Sustainable Products Regulation (ESPR) now mandates Digital Product Passports (DPPs) —standardised lifecycle records—for most EU products [7]. With the implementation of compliance in 2027, reliable DPP solutions have become a market urgent need[7].

Our product EcoPassEU is a blockchain-based DPP platform which allows manufacturers to record product information as permanent, publicly verifiable entries on the Ethereum blockchain. Unlike traditional databases, the immutability of blockchain records means that once the data is released, it cannot be altered retrospectively, eliminating the possibility of post-hoc falsification [5]. Any user can obtain a complete and encrypted product history by entering the product ID or scanning the QR code.

EcoPassEU serves four key stakeholders. It provides SME manufacturers with a cost-effective ESPR compliance channel through a web for batch data entry and automated QR code generation. At the same time, consumers, regulators and retailers can also get open and wallet-free access. Just search for the product ID to immediately obtain the blockchain-verified product passport and supply chain map, enabling the independent verification of environmental claims without prior technical expertise.

The value of EcoPassEU is threefold. Primarily, it provides manufacturers with a practical way to ESPR deadlines, reducing the risk of market exclusion. Meanwhile, the platform directly cracked down on greenwashing, securing product records via cryptographic hashing replaces brand self-reporting with independent, third-party verification. Thirdly, it promotes the development of circular economy by equipping recyclers with the precise material data necessary to recover resources when products are scrapped.

2 Background

2.1 Digital Product Passports and the Regulatory Context

A Digital Product Passport (DPP) is a structured digital record encompassing lifecycle data—such as material composition, provenance, and end-of-life guidance—designed to inform stakeholders and support more sustainable design choices, efficient resource use, and the transition towards circular business models [1,2,4,13].

The regulatory foundation for DPPs comes from the European Green Deal and Circular Economy Action Plan [10,11]. Based on these 2 frameworks, the Ecodesign for Sustainable Products Regulation (ESPR), came into force on 18 July 2024,, mandates DPPs for almost all physical goods sold in the EU, regardless of the seller’s location [7].

2.2 The Problem: Trust in Sustainability Claims

Despite regulatory momentum, the implementations of most DPPs rely on centralised, proprietary databases [17]. This architecture encourages the selective concealment of data, hindering aftermarket participants like independent repairers [8]. Furthermore, managing huge volume of lifecycle data centrally raises significant concerns regarding storage efficiency, data integrity, and vulnerability to breaches [21]. Taken together, downstream actors — including importers, recyclers, and end consumers — lack reliable ways to independently verify product information.

This problem is exacerbated by the prevalence of greenwashing, broadly understood as false, vague, or unsubstantiated environmental claims that make products appear more sustainable than they are [14]. A 2020–2021 EU sweep of 344 online claims found that nearly half were potentially deceptive—often substituting verifiable details with vague terms like “conscious”—and 59% lacked accessible evidence [18]. Another EU website screening found that 42% of sustainability claims were misleading enough to constitute improper commercial behaviour [18].

Blockchain provides a structural solution to this trust problem. Because records on a public blockchain are replicated across a distributed network, neither single party can unilaterally tamper with the records, providing a form of tamper evidence that cannot be replicated by a centralised system [5,16]. Smart contract introduced in the Ethereum protocol allows data access and update rules to be encoded in publicly readable logic, so that there is no need for trusted intermediaries and enabling cross-organisational verification without shared infrastructure [23].

2.3 Related Work and Existing Platforms

Blockchain-based product traceability uses its immutability and decentralised verification to improve supply chain transparency [5]. Blockchain-based product traceability emerged as a research topic around 2015–2016, with early work

concentrated in food and agriculture before spreading to pharmaceuticals, electronics, and textiles [9,19]. More recently, research has shifted from feasibility towards scale and efficiency: for instance, Walmart’s use of IBM Food Trust — built on Hyperledger Fabric — to reduce food-item tracing time from six days to under three seconds [9] .

In the commercial field, Circularise is among the closest comparable platforms, offering DPPs with selective disclosure of sustainability data across supply chains, with applications spanning the automotive, battery, and textile sectors [6]. VeChain similarly deploys blockchain and smart contracts on its VeChainThor platform to achieve supply chain transparency, though it runs on a permissioned network with a restricted set of authority-approved validators [20] Both platforms have a common limitation: reliance on proprietary infrastructure or permissioned networks introduces vendor dependency and reduces interoperability [3,22].

EcoPassEU takes a different approach. Rather than relying on a proprietary database, it is built on the public Ethereum blockchain and IPFS, with all rules encoded in open smart contracts. Product data is stored on IPFS, whilst only a small set of cryptographic identifiers is written on-chain, keeping costs low. The batch registration function further reduces the cost per product, making the platform viable for SMEs. And as an open-source project, it carries no vendor lock-in risk.

2.4 Business Viability

EcoPassEU’s revenue model rests on two foundations. The first is public funding: the EU’s CIRPASS-2 project (2024–2027) explicitly subsidise DPP deployment across textiles, electronics, and construction, with dedicated support for SMEs adopting “DPP-as-a-Service” models [12]. As an open source platform, EcoPassEU is fully capable of benefitting from such grants.

Another is subscription revenue. ESPR compliance is legally mandatory for any business selling physical goods in the EU, forming a captive market independent of voluntary commitments [7] The green technology and sustainability market is projected to grow from USD 25.47 billion in 2025 to USD 73.90 billion by 2030 [15]. EcoPassEU charges a graded subscription fee based on passport volume, with optional premium tiers for API integration and compliance reporting. The platform is built on public Ethereum, supporting payment with ETH or other ERC-20 tokens, providing international users a lower-friction settlement option and removing the need for traditional banking intermediaries.

3 Design Documentation

Outline the system architecture of EcoPassEU. Include your justifications for using EVM-compatible networks like Polygon, ERC-1155 tokens, IPFS, and DIDs/VCs. Insert your Use Case diagrams, Sequence/Flow diagrams, and front-end wireframes here.

4 Project Execution

Detail the implementation process of the DApp. This should feature core code analysis, data structures, your testing strategy, final UI screenshots, and a brief user manual on how to interact with the application.

4.1 Critical Evaluation

Provide a technical analysis of EcoPassEU, evaluating the gas usage of your smart contracts, security considerations (especially regarding the separation of on-chain and off-chain data), and runtime performance.

5 Conclusion

Summarize the main theoretical and technical findings. Detail what the group achieved, acknowledge limitations (e.g., the physical-to-digital bridge vulnerabilities), and propose future work.

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A Appendix

This appendix includes the mandatory materials required for the FTGP summative assessment.

A.1 Smart Contract Deployment

- **Smart Contract Address(es):** [Insert Sepolia testnet addresses here]
- **Etherscan Links:** [Insert links to verified contracts on Etherscan]

A.2 Project Links

- **Deployed Website:** [Insert URL]
- **GitHub Repository:** https://github.com/DeTaiDong/FTGP2425_Group1_11

A.3 Transaction Hashes

- **Deployment TX:** [Insert Hash] - Explanation.

A.4 Equity Shares & Contributions

- Luxiao Cao: [Weight]
- Akshansh Rajora: [Weight]
- Fuyu Cao: [Weight]
- Tianwei Yu: [Weight]
- Detai Dong: [Weight]

[Insert Screenshot of GitHub - Insights - Contributor here]

A.5 JIRA Project Board

[Insert screenshots of 'Summary', 'timeline', and 'backlog' pages here]

A.6 Individual Reflection Essays

Luxiao Cao [Insert 500-word reflection here]

Akshansh Rajora [Insert 500-word reflection here]

Fuyu Cao [Insert 500-word reflection here]

Tianwei Yu [Insert 500-word reflection here]

Detai Dong [Insert 500-word reflection here]

A.7 Team Sprint Reports

[Insert chronological sprint reports here]

A.8 Complete Source Code

Smart Contracts

```
// Insert complete Solidity code here
```

Frontend Core Files

```
// Insert modified/relevant frontend files (e.g., page.js) here
```